CYBER FORENSIC DEPARTMENT

DIGITAL FORENSICS & INCIDENT RESPONSE (DFIR) UNIT

FORENSIC INVESTIGATION REPORT

INCIDENT ASSESSMENT: BENIGN / NORMAL ACTIVITY

RESTRICTED - LAW ENFORCEMENT & SOC CONFIDENTIAL - CHAIN OF CUSTODY PRESERVED

Case Identifier:	CASE-0002	**Report Version:**	v2.5 Enterprise DFIR
Target Evidence:	`attack.txt`	**Report Date:**	24-Sep-2026 16:33:18 UTC+05:30
Incident Severity:	**MEDIUM (49/100)**	**Investigation Status:**	Active Containment / Triage Complete
Primary Analysts:	Pragati Patil, Tanisha Lohar, Gaurav Rajguru	**Project Guide:**	Ms. Tejas P. Shinde
Department:	Department of Computer Science and Engineering (IoT & Cyber Security including Blockchain)	**Institution:**	ADCET, Ashta (Autonomous)

1. EXECUTIVE SUMMARY

The Cyber Forensic Department initiated an automated digital forensic triage of the evidence artifact **'attack.txt'** (139 bytes) to ascertain unauthorized activity, malicious payload execution, lateral movement, and data exfiltration. The automated triage engine classified the incident with an overall risk severity rating of **MEDIUM (Risk Score: 49/100)**.

Supervised Machine Learning classification (Random Forest) identified the primary threat vector as **'Benign / Normal Activity'** with **100.0% confidence**. The attack has advanced to Cyber Kill Chain stage: **'Critical Stage - Active Threat Impact / Ransomware Execution'** (100.0% progression). Unsupervised Anomaly Modeling (Isolation Forest) confirmed system deviation status as **'Normal'** (Anomaly Index: **0.00**).

Analysis discovered a cumulative total of **0 Indicators of Compromise (IOCs)**, **0 heuristic YARA signature matches**, and **3 correlated MITRE ATT&CK; techniques**. Immediate endpoint containment, credential revocation, and perimeter firewall blocking are required.

2. SCOPE AND METHODOLOGY

The scope of this forensic triage encompasses the examination of digital evidence submitted to the Cyber Triage platform. The analysis followed a forensically sound methodology compliant with ISO/IEC 27037 standards for digital evidence handling:

Evidence Artifact	SHA-256 Cryptographic Hash	Size	Format / Status
`attack.txt`	`7e18f06d1c29e60afd0007cbb45e9120287728b001ce51bc7160fb36d2dc5e01`	139 B	TXT (Preserved)

- **Cryptographic Verification:** Bit-level hashing (SHA-256, MD5, SHA-1) upon intake to ensure zero evidence tampering.
- **Pattern & IOC Extraction:** Automated extraction of IPv4/IPv6, URLs, domains, hashes, CVEs, and Base64 payloads.
- **YARA Heuristic Scanning:** Heuristic detection of ransomware strings, shadow copy deletion, Mimikatz, and webshells.
- **Machine Learning Modeling:** Dual AI architecture combining Isolation Forest anomaly scoring with Random Forest classification.
- **Threat Intelligence Enrichment:** Live reputation correlation against known threat actor infrastructures and C2 gateways.
- **ATT&CK; & Kill Chain Mapping:** Normalization of forensic artifacts into MITRE Enterprise Tactics and Kill Chain stages.

3. AI ANOMALY & THREAT CLASSIFICATION RESULTS

Analytical Metric	Quantitative Value	Forensic Interpretation

Overall Threat Risk Score	**49 / 100**	Calibrated severity level: MEDIUM
Supervised Threat Classifier (RF)	**Benign / Normal Activity**	Multi-class probability distribution: Benign / Normal Activity: 100.0%
Isolation Forest Anomaly Index	**0.00 / 1.00**	Unsupervised deviation status: Normal
Cyber Kill Chain Progression	**100.0%**	Lifecycle stage reached: Critical Stage - Active Threat Impact / Ransomware Execution
Shannon Binary Entropy	**4.40 / 8.00**	Measures code randomness, packing, or encryption density

4. MITRE ATT&CK; ENTERPRISE MAPPING

Tactic Name	Technique ID	Technique Name	Observed Artifact / Trigger
Impact	**T1486**	Data Encrypted for Impact	`ransomware`
Initial Access	**T1190**	Exploit Public-Facing Application	`rce`
Credential Access	**T1110**	Brute Force: Password Guessing	`brute force`

5. INDICATORS OF COMPROMISE (IOCs)

5.1 Network Indicators (Defanged)

No external network indicators detected.

5.2 Host & Endpoint Indicators

No suspicious host-based indicators detected.

6. CHRONOLOGICAL INCIDENT TIMELINE

Timestamp (UTC)	Event Description	Category	Severity
`Unspecified Timestamp`	ransomware	Ransomware / Critical Threat	**Critical**
`Unspecified Timestamp`	failed login	Authentication Failure	**High**
`Unspecified Timestamp`	brute force	Authentication Failure	**High**

7. YARA & HEURISTIC DETECTION SUMMARY

No signature or YARA-style heuristic rules matched.

8. CONTAINMENT, ERADICATION & RECOMMENDATIONS

8.1 Immediate Containment Actions (0–24 Hours):

8.1.1 [Identity Protection] - Force password reset and revoke active Kerberos TGT tickets for all privileged and administrative accounts.

8.1.2 [MFA Enforcement] - Enforce Multi-Factor Authentication (MFA) across all remote access and administrative portals.

8.2 Short-Term Eradication (1–7 Days):

• Terminate unauthorized persistence mechanisms (Registry Run keys, scheduled tasks, and rogue services).

• Invalidate all compromised administrative credentials and force enterprise-wide Kerberos ticket resets.

• Re-image infected hosts using verified, immutable baseline golden images.

8.3 Long-Term Hardening & Post-Incident Security:

- Enforce hardware-backed Multi-Factor Authentication (MFA) across all administrative access points.
- Deploy Endpoint Detection & Response (EDR) agents with tamper protection and automated containment.
- Implement air-gapped, immutable offline backups to prevent ransomware impact.

9. CHAIN OF CUSTODY SUMMARY & CRYPTOGRAPHIC VERIFICATION

Timestamp	Action Performed	Performed By	Verification Hash / Block
`2026-08-20 21:32:34`	Evidence Acquisition & Cryptographic Hashing	Gaurav Rajguru / Pragati Patil / Tanisha Lohar	`SHA256: 7e18f06d1c29e60afd0007cbb45e...`
`24-Sep-2026 16:33:18 UTC+05:30`	AI Forensic Triage & Multi-Parser Extraction	Cyber Triage AI Core	**Hash Chain: VERIFIED INTACT**

10. LIMITATIONS AND ASSUMPTIONS

1. **Evidence Completeness:** Findings are strictly bounded by the digital artifacts provided for triage. Unacquired physical RAM or unallocated disk sectors may harbor auxiliary attacker footprints.
2. **Obfuscation & Encryption:** Adversary anti-forensic techniques (e.g. encrypted payloads, memory unhooking) may limit static string detection.
3. **Legal Admissibility:** Cryptographic hashes and custody logs have been maintained in compliance with Section 65B of the Indian Evidence Act.

11. AUTHORISATION & FORMAL SIGN-OFF

The undersigned forensic examiners and reviewing authorities hereby certify that this investigation was conducted under rigorous forensic protocols, and the findings presented herein represent an accurate, tamper-proof analysis of the evidence.

Role	Name & Academic / Professional Designation	Signature & Official Stamp	Date
Lead Forensic Examiners	**Pragati Kiran Patil (23101024)** **Tanisha Prakash Lohar (23101009)** **Gaurav Subhash Rajguru (23101004)** B.Tech CSE (IoT, Cyber Security & Blockchain)	_____	24-09-2026
Reviewing Officer & Guide	**Ms. Tejas P. Shinde** Guide, Dept. of CSE (IoT, CS & BT), ADCET Ashta	_____	24-09-2026
Department Coordinator	**Mr. Samish N. Kamble** Project Coordinator, CSE [IOT-CSBT], ADCET Ashta	_____	24-09-2026
Head of Department	**Dr. Tahseen A. Mulla** Head of Department, CSE [IOT-CSBT], ADCET Ashta	_____	24-09-2026